D4 — Deployment Demonstration and Dashboard

Boréal Marché Customer Intelligence Platform · Capstone Project 1

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1. Live system and evidence index

Item	Where	Status
Application (5 sections, EN/FR toggle)	**PENDING**	App Runner service `boreal-ui`
API documentation (OpenAPI)	**PENDING**	App Runner service `boreal-api`
`/version` = deployed commit	**PENDING**	asserted by the CD smoke test (§5)
Screen recording, 3–5 min (prediction, recommendation, assistant)	**PENDING**	link also in `README.md`
CI: pull request with all checks green; workflow file	PR `feat/d4` → `master`; `.github/workflows/ci.yml`	§4
CD: merge-triggered deploy, SHA-tagged image, smoke test; OIDC	Actions run of `.github/workflows/cd.yml`	§5
Monitoring dashboard with shifted cohort and triggered alerts	*Monitoring* tab; evidence in §7	measured locally on the release image

Items marked PENDING are produced by the one-time AWS bootstrap and the first merge (`docs/runbooks/AWS_APPRUNNER_DEPLOY.md`); this note is regenerated from the evidence files once they exist (`python scripts/collect_d4_evidence.py --api-url … --ui-url … --video-url …`, then `python -m boreal_marche.reporting.build D4`). Nothing below is typed in by hand.

2. What is deployed

One API container serves all three heads and the monitoring endpoints; one interface container talks to it over HTTP only. Offline artifacts (model, rules, retrieval index, reference profile) are baked into the API image at build time, so a deployed version is exactly one image digest.

Endpoint	Purpose
`/health`, `/ready`	liveness (process up) vs readiness (model, rules, index, reference loaded; 503 otherwise)
`/version`	API version, deployed commit, image tag, model version and source, rules mining date
`/predict`, `/predict/batch`	Head A — calibrated P(order within 90 days) and the *proposed* coupon decision
`/recommend`, `/products`	Head B — rules indexed by antecedent, strategy that fired; catalogue search
`/customers`, `/customers/{ref}/features`	pseudonymised demo customers: the feature-store stand-in
`/chat`, `/chat/stream`, `/chat/providers`	Head C — grounded assistant (JSON or Server-Sent Events)
`/metrics`, `/monitoring/report`, `/monitoring/alerts`, `/monitoring/timeseries`	Prometheus metrics; drift/quality/ops report with traffic lights; alert history

Every request model is Pydantic v2 with `extra="forbid"`; errors are typed (422 validation, 404 unknown customer, 503 artifact or provider unavailable, 500 with a request id — never a bare stack trace). Models, rules and the index load once in the lifespan handler. Every request is written to `request_log` and as a JSON log line; every prediction (full feature vector), recommendation and chat turn to its own table — the dashboard is built from these rows. **Tests:** 80 pass, line coverage 84% (CI gate 70%), including the API contract suite and the golden-prediction regression test, which the post-deploy smoke test repeats against the live service.

Head C, built for this deliverable. 6 policy documents (4,995 words: delivery, returns, allergens, loyalty, privacy in EN+FR, ordering) plus the catalogue export, chunked by section into 66 chunks with metadata and embedded with `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` (ONNX, no torch), so French questions retrieve English chunks. Four typed tools — `policy_search`, `retention_score` and `recommendations` (which call this API's own `/predict` and `/recommend` over HTTP) and `product_lookup` — plus an explicit `out_of_scope` choice. Guardrails run around every provider: another customer's data is refused before any model runs; `retention_score` has no customer argument at all; a policy answer below similarity 0.30 is handed to a human; every grounded answer carries citations; an LLM answer that mentions a SKU or percentage absent from the tool output is replaced by the verified template.

Question set	Provider	Tool choice	Source cited	OOS refused	False refusals	Latency p50	Cost/query
golden (40)	rule	100.0%	95.0%	100.0%	0.0%	4 ms	$0.0000
golden (40)	ollama	97.5%	90.0%	100.0%	3.2%	1553 ms	$0.0000
golden	openai	not run: OPENAI_API_KEY not set					
paraphrase (20)	rule	35.0%	22.2%	80.0%	73.3%	0 ms	$0.0000
paraphrase (20)	ollama	95.0%	88.9%	100.0%	6.7%	1282 ms	$0.0000
paraphrase	openai	not run: OPENAI_API_KEY not set					

Table 1 — Assistant evaluation (`reports/metrics/assistant_eval.json`). The golden set (40 questions) was written before the assistant; the paraphrase set (20) after the rule provider's keyword lists, to measure their overfitting. OpenAI is implemented behind the same interface but not evaluated: no API key (course §8.5 also forbids sending personal data to it).

The rule baseline scores 100% on the golden set and 35% on paraphrases: its keywords were written by someone who had read the golden questions. The local model (Ollama, llama3.1:8b) holds up — 97.5% and 95% — at ~1.5 s per answer and $0. That is the measured case for the language model: routing natural phrasings and answering French questions from English sources. The rule-tool-choice check on the golden set runs in CI.

3. Containers

Image	Unpacked size	Compressed (pushed to ECR)	Contents
API v0.3 (D3, single venv, `pip install .`)	1437 MB	719 MB	Heads A+B only
API v0.9 (this Dockerfile)	823 MB	407 MB	Heads A+B+C, monitoring, 241 MB embedding model baked in
Interface v0.9	500 MB	155 MB	Streamlit + Altair, no model code

Table 2 — Image sizes measured on the built images (`reports/metrics/d4_evidence.json`).

The API image is **43% smaller** than the D3 image while now also containing the assistant and its embedding model. Design choices in `docker/Dockerfile`:

Choice	Why
Base `python:3.11-slim` **pinned by digest**	a tag can be re-pointed; a digest makes the OS layer reproducible
Multi-stage: `builder` → `runtime`	pip caches, build tools and wheel test suites stay in the builder
Serving lock `docker/requirements-api.txt`	only what the API imports; `xgboost-cpu` drops ~200 MB of CUDA/NCCL wheels; no pyarrow, MLflow, Optuna, matplotlib
Dependencies copied before source	a code change rebuilds one small layer, not 60 wheels
Embedding model downloaded at build time, `HF_HUB_OFFLINE=1`	no network at start-up; cold start does not depend on Hugging Face
Non-root user UID 10001	a compromised process cannot write the image; CI asserts the UID
`HEALTHCHECK` on `/ready`, not `/health`	healthy only once every artifact is loaded
`GIT_SHA` / `IMAGE_TAG` as the last layer	`/version` proves which commit runs; changing it invalidates nothing else
Allow-list `.dockerignore` (deny `*`, allow 13 paths)	data, notebooks, MLflow state, `.env` can never reach the build context

`docker compose -f docker/compose.yml up --build -d` starts API, interface, Postgres (prediction log) and Ollama with a one-shot model pull. I tested API + interface + Postgres from the images (7/7 smoke checks, predictions persisted in Postgres); the Ollama container's 4.9 GB model pull was not repeated in the test (the evaluation in Table 1 used the same model via a host install).

4. Continuous integration

`.github/workflows/ci.yml` runs on every pull request: **ruff** (lint + format), **mypy --strict**, **pytest with a 70% coverage gate** (unit, API contract, golden prediction, monitoring, assistant guardrails), the **data-quality suite**, the **assistant tool-choice evaluation** on the golden set, a **Docker build of both images with their sizes in the job summary**, a **container smoke test** (runs the API image, checks UID 10001, the health status and the same 6 API checks as CD), and **gitleaks** over the full history. Branch protection on `master` requires the three jobs and a pull request. Retrieval in CI uses a deterministic hashing embedder (`BOREAL_EMBEDDER=hash`) so tests need no model download; retrieval *quality* is measured by the evaluation in Table 1, not by CI.

5. Continuous deployment and OIDC

On merge to `master`, `.github/workflows/cd.yml`: (1) requests a GitHub OIDC token and exchanges it for one-hour credentials of role `BorealGithubDeploy` (`aws-actions/configure-aws-credentials`); (2) prints the assumed identity; (3) builds both images tagged with the **commit SHA** and pushes them to ECR (immutable tags, scan on push); (4) points the two App Runner services at the new tags (`scripts/apprunner_deploy.py`) and waits for success; (5) runs `scripts/smoke_test.py`, which fails the job unless `/version` equals the merged commit, the golden prediction is reproduced, a rule fires and the assistant cites a source.

Why OIDC. No AWS key exists in GitHub: the role's trust policy (`infra/aws/github-oidc-trust.json`) accepts only tokens whose subject is `repo:OlgaMorgunova/boreal-marche-capstone:environment:production`, and the environment is restricted to `master`. Its permissions (`infra/aws/github-deploy-policy.json`) are push to two repositories, update/describe two services, pass one role to App Runner. A leaked log or fork cannot deploy, and nothing needs rotating.

Scheduled retraining. `.github/workflows/retrain.yml` (Mondays) re-runs ETL, retrains, re-mines rules, logs to MLflow, and evaluates the promotion policy against the committed champion on the validation snapshot — now including a **fairness gate** (subgroup lapser recall by region, tenure band and spend quartile; `boreal_marche.retention.promote`). Only if every gate passes does it open a pull request; it never deploys. A dry run with the extra-trees challenger correctly returned "no promotion". The same gate exposed a real gap that D5 must address: lapsers in the top spend quartile receive the offer far less often than average (documented in `docs/MLFLOW_PROMOTION_POLICY.md`).

6. Hosting and cost

App Runner was chosen over Render because it deploys straight from ECR with the OIDC role and needs no second account. **Region deviation:** App Runner is not offered in `ca-central-1` (console message, checked 2026-10-24), so ECR and App Runner run in `us-east-1`, the closest supported region; the SageMaker job stays in `ca-central-1`. This is acceptable for the demonstration because the deployed service only processes the public, pseudonymised UCI data. For real customers it would be a transfer of personal information outside Québec (Law 25, s. 17: prior assessment, adequate protection), and I would deploy the same image to ECS Fargate behind an Application Load Balancer in `ca-central-1` (≈ $30/month more, mostly the load balancer) — the cost of data residency, stated rather than hidden. The API needs ~750 MiB at idle (measured with `docker stats`), so it runs at 1 vCPU / 2 GB; the interface at 0.25 vCPU / 0.5 GB. Estimated ≈ $13/month while running, $0 while paused; both services are paused between grading windows (runbook §4). Single instance by design: the SQLite prediction log lives on the instance and is lost on redeploy — the Compose stack shows the production design (Postgres). Ollama is not deployed (an 8B model needs a GPU instance); the live assistant offers the rule provider.

7. Monitoring with simulated and shifted traffic

The traffic simulator is a **Locust** file adapted from the course's Locust lab (`loadtest/locustfile.py`: `HttpUser`, weighted tasks, `catch_response`, grouped names). Phase 1 replays the 300 held-out demo customers (1821 requests, 0 failures, 15.3 req/s); phase 2 sends a deliberately shifted cohort — recent Maritimes sign-ups with one or two orders, no cadence history, lower spend, and baskets of SKUs the rules have never seen (1817 requests, 0 failures). Both phases ran against the release image; the report was saved after each (`scripts/simulate_traffic.py`).

Check	After normal traffic	After shifted cohort	Measure
input drift	amber (0.000)	red (0.528)	share of features with PSI >= red
prediction drift	green (0.032)	red (2.112)	PSI of p_return
data quality	green (0.042)	red (0.994)	max(missing-rate change, /predict 422 rate)
error rate	green (0.000)	green (0.000)	share of 5xx responses
latency p95 predict	green (67.317)	green (70.200)	ms, server-side
rule fire rate	green (1.000)	red (0.188)	share of non-empty baskets answered by a rule

Table 3 — Traffic lights after each phase (`reports/metrics/monitoring_*.json`). Thresholds: `params.yaml → monitoring`, explained in `docs/MONITORING.md`.

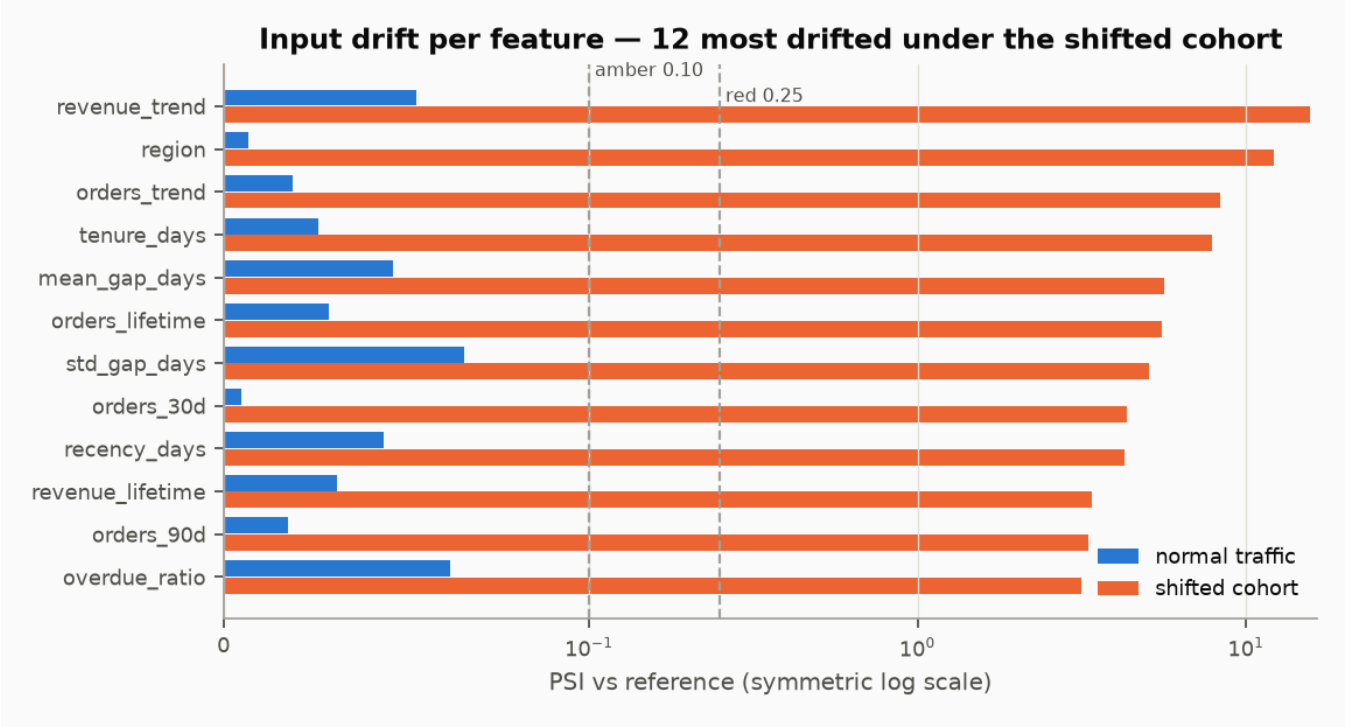


Figure 1 — Input drift per feature, normal vs shifted traffic (top drifted: `revenue_trend` (PSI 15.63), `region` (PSI 12.12), `orders_trend` (PSI 8.35), `tenure_days` (PSI 7.84)).

Under the shifted cohort 53% of features cross PSI 0.25, 85% of basket items are unknown and the rule fire-rate falls to 19% (reference 92%). Normal traffic is not perfectly green: one or two features sit just above PSI 0.10 (amber) with 500 predictions from only 300 customers — sampling noise I report rather than tune away. Each level change raised an alert (7 in total), written to `alert_log`, logged as a JSON `alert` line, posted to `ALERT_WEBHOOK_URL` when set, and — for the deployed service — turned into a GitHub issue by the hourly `monitor.yml`:

Time (UTC)	Check	Transition	Value	Notified via
13:31:29	data_quality	green → **amber**	0.062	log
13:32:35	data_quality	amber → **green**	0.032	log
13:32:36	input_drift	green → **amber**	0.000	log
13:33:30	input_drift	amber → **red**	0.528	log
13:33:30	prediction_drift	green → **red**	2.122	log
13:33:30	data_quality	green → **red**	0.996	log
13:34:30	rule_fire_rate	green → **red**	0.200	log

A design correction found by monitoring. I first used the pooled training snapshots as the drift reference. Healthy next-quarter traffic was then permanently red: 8% of features had PSI ≥ 0.25 (`tenure_days` 1.15, `std_gap_days` 0.30, `orders_lifetime` 0.26), because the data start in December 2009, so every later snapshot has a longer observable history and lifetime features grow by construction. A monitor that is always red is ignored. The reference is now the calibration snapshot (2,358 rows, demo customers held out), and the training-vs-reference shift is recorded in the reference profile instead of alerted on.

Delayed ground truth. The label exists 90 days after a prediction. A weekly backfill joins predictions older than 90 days with the customer's orders and writes `ground_truth`; the dashboard prints the date of the first true accuracy signal (first prediction + 90 days + up to 7 days of job cadence). Replaying the known historical outcomes of the demo customers through this pipeline labelled 2,849 logged predictions (realised PR-AUC 0.768, lapser recall 0.775); these labels are in-sample for the calibrator, so they prove the pipeline, not a new performance estimate.

Latency. Server-side `/predict` p95 was 70 ms through Docker Desktop's port proxy on Windows, but 13.5 ms (and `/recommend` 2.2 ms) when called from inside the Docker network: the gap is the proxy delaying POST bodies, not the model. The latency profile by endpoint is D5 work.

8. Reproducibility statement (a stranger's commands)

```
git clone https://github.com/OlgaMorgunova/boreal-marche-capstone.git && cd boreal-marche-capstone
docker compose -f docker/compose.yml up --build -d      # API :8000/docs, UI :8501, Postgres, Ollama
python3 scripts/smoke_test.py --api http://127.0.0.1:8000 --ui http://127.0.0.1:8501
pip install "locust>=2.30" && python3 scripts/simulate_traffic.py --seconds 120   # fills the dashboard
```

Without Docker (Windows: `docs/RUN_FROM_ZERO_WINDOWS.md`): `make setup-pinned`, `make api`, `make frontend`, `make simulate`. Rebuilding every artifact from the raw data: `make all` (≈ 15 min). No secret is needed for any of these; optional variables are listed in `.env.example`.

9. Limitations and what is pending

Live URLs, the CI/CD run screenshots and the video depend on the AWS bootstrap and the first merge (PENDING above). App Runner's prediction log is per-instance and ephemeral. The deployed assistant has no LLM provider. The Ollama Compose service was not tested end to end. The SageMaker HPO job is still waiting for the spot-instance quota increase (case opened 2026-09-29); D3 §11 is updated when it runs. The fairness gap in the top spend quartile is unresolved (D5).

10. Sources and AI-tool use

Course Locust lab (instructor material) for the load-test pattern; AWS documentation for App Runner, ECR and GitHub OIDC federation; Health Canada and Law 25 summaries for the fictional corpus (to be verified against primary sources in D5). AI-assisted coding and drafting tools were used for code, tests, the policy corpus and this text; I reviewed each module and can explain every line. **Where AI-generated output had to be corrected:** the drift reference (pooled training data made healthy traffic red, §7); a test that hid the SKU guardrail by stripping spaces from streamed tokens; the Docker clean-up step that deleted `numpy/_core/tests` and crashed the container at start-up (caught by the container smoke test); and the French detector, which missed accent-free questions.